

Math 340: Elementary Matrix and Linear Algebra

FINAL EXAM

Sunday, May 5th, 2024, 7:45am-9:45am

Please circle your Instructor and your Discussion Session (date and start time given) in the table below.

Dr. Péter Juhász	Dr. Sean Paul	Dr. Terry Harris
Awildo Gutierrez, M 7:45 AM	Jiaming Xu, T 11:00 AM	Ryan Tamura, T 12:05 PM
Awildo Gutierrez, M 8:50 AM	Jiaming Xu, T 12:05 PM	Rune Chen, T 1:20 PM
Hongyu Zhu, W 9:55 AM	Jiaming Xu, R 11:00 AM	Ryan Tamura, R 12:05 PM
Awildo Gutierrez, W 7:45 AM	Zaidan Wu, R 12:05 PM	Hanzhang Mao, R 1:20 PM
Hongyu Zhu, W 8:50 AM	Zaidan Wu, R 1:20 PM	Connor Simpson, T 2:25 PM
Jacob Denson, M 9:55 AM		Ryan Tamura, T 11:00 AM
Carsen Grote, M 8:50 AM		Lauren Neudorf, R 2:25 PM
Carsen Grote, M 9:55 AM		Connor Simpson, T 12:05 PM
Jacob Denson, W 8:50 AM		Hanzhang Mao, R 12:05 PM
Carsen Grote, W 9:55 AM		Connor Simpson, T 9:55 AM
Jacob Denson, W 9:55 AM		

Name: Nicolas Bourbaki Wisc email: _____

I pledge that the work on this exam is entirely my own. I understand that the penalties for cheating may include an F in the course and referral to my dean for further action.

Student Signature: _____

READ THE FOLLOWING INFORMATION.

- This is a 120-minute exam. It consists of 9 problems for a total of 100 points; the exam is 7 sheets of paper, including this cover sheet. It is your responsibility to make sure that you have a complete exam.
- Books, notes, calculators, and other aids are not allowed.
- Problems are spaced out to allow ample room for work. You may use the last page for scratch work, but it will not be graded. Do not unstaple or remove pages as they can be lost in the grading process. **An incomplete exam packet will result in an automatic zero.**
- Only complete, well written, and neat solutions will be awarded full credit. Remember that all claims must be supported. Responses which do not meet these qualifications may be awarded some partial credit.
- Notation reminders: M_{mn} is the vector space of $m \times n$ real matrices with standard matrix addition and scalar multiplication, P_d is the vector space of polynomials (in x) of degree at most d with standard polynomial addition and scalar multiplication, $\mathbf{0}$ denotes the zero vector in a vector space V . $\mathbb{R}^n$ uses the standard vector addition/scalar multiplication unless otherwise noted.

DO NOT BEGIN THIS EXAM UNTIL SIGNALLED TO DO SO.

1. (12 points, 2 points each) Are each of the following true or false statements? Fill in the bubble next to the correct answer. Justification is not necessary.

a.) If $T : P_2 \rightarrow \mathbb{R}^3$ is a linear transformation, then T is an isomorphism.

- ☐ True
☒ False

may not be! ex: $T(p) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

b.) P_2 and $\mathbb{R}^3$ are isomorphic.

- ☒ True
☐ False

c.) The following function $\left\langle \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}, \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} \right\rangle = x_1 y_1 - x_2 y_2$ defines an inner product on $\mathbb{R}^2$.

- ☐ True
☒ False

$\left\langle \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\rangle = -1 < 0 \times$.

d.) Suppose $B = (b_1, b_2, b_3)$ and $D = (d_1, d_2, d_3)$ are two ordered bases for V . If $C_B(\mathbf{v}) = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

and $C_D(\mathbf{w}) = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, then $\mathbf{v} = \mathbf{w}$.

- ☐ True
☒ False

ex: $V = P_2$ $B = (1, x, x^2)$, $D = (1, x^2, x)$
 $C_B(1 + 2x + 3x^2) = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = C_D(1 + 2x^2 + 3x)$
 not equal

e.) If V and W are subspaces of P_4 and $\dim V = \dim W = 4$, then $V = W$.

- ☐ True
☒ False

ex: $V = \text{span}\{1, x, x^2, x^3\}$
 $W = \text{span}\{x, x^2, x^3, x^4\}$

f.) Suppose $M = \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix}$ and $N = \begin{bmatrix} 5 & 1 \\ -1 & 2 \end{bmatrix}$. M and N can be matrix representations for the same linear operator $T : P_1 \rightarrow P_1$ using different bases. similar.

- ☐ True
☒ False

$\text{tr}(M) = 4 \neq 7 = \text{tr}(N)$.

2. (8 points) Consider $\mathbb{R}^4$ with the standard inner product (also known as the dot product), and consider the set W of vectors which are orthogonal to both $\begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \end{bmatrix}$ AND $\begin{bmatrix} 3 \\ 5 \\ -1 \\ 2 \end{bmatrix}$. Find a basis for W .

Show all work. (You may assume that W is a subspace of $\mathbb{R}^4$.)

$$\text{Let } \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \text{ be in } W.$$

$$\begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \end{bmatrix} \cdot \vec{x} = x_1 + 2x_2 + x_3 = 0$$

$$\begin{bmatrix} 3 \\ 5 \\ -1 \\ 2 \end{bmatrix} \cdot \vec{x} = 3x_1 + 5x_2 - x_3 + 2x_4 = 0$$

$$\text{sys: } \left[\begin{array}{cccc|c} 1 & 2 & 1 & 0 & 0 \\ 3 & 5 & -1 & 2 & 0 \end{array} \right] \xrightarrow{-3r_1+r_2} \left[\begin{array}{cccc|c} 1 & 2 & 1 & 0 & 0 \\ 0 & -1 & -4 & 2 & 0 \end{array} \right]$$

$$\downarrow \begin{array}{l} -r_2 \rightarrow r_2, \\ -2r_2 + r_1 \rightarrow r_1 \end{array}$$

$$\left[\begin{array}{cccc|c} 1 & 0 & -7 & 4 & 0 \\ 0 & 1 & 4 & -2 & 0 \end{array} \right]$$

$$x_3 = r$$

$$x_4 = t$$

$$x_1 = 7r - 4t$$

$$x_2 = -4r + 2t$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 7r - 4t \\ -4r + 2t \\ r \\ t \end{bmatrix} = r \begin{bmatrix} 7 \\ -4 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -4 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{Basis: } \begin{bmatrix} 7 \\ -4 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -4 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

3. (18 points total) (This problem continues onto the next page)

Let $L : P_1 \rightarrow P_1$ be a linear transformation with $M_B(L) = \begin{bmatrix} 2 & -3 \\ -4 & 6 \end{bmatrix}$ with respect to the ordered basis $B = (x+1, x-2)$.

(NOTE: B is NOT the standard basis for P_1).

$\parallel$
 A

a. (4 points) Compute $L(4x+1)$. Show all work.

(Note: your final answer should be a polynomial, not a column vector).

$$4x+1 = a(x+1) + b(x-2)$$

$$\begin{array}{r} a+b=4 \\ -(a-2b=1) \\ \hline 3b=3 \\ b=1 \\ a=3 \end{array}$$

$$\begin{bmatrix} 2 & -3 \\ -4 & 6 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ -6 \end{bmatrix}$$

$$\boxed{3(x+1) - 6(x-2)}$$

b. (4 points) Is L diagonalizable? Justify your answer.

$$\det(\lambda I - A) = \det \begin{bmatrix} \lambda-2 & 3 \\ 4 & \lambda-6 \end{bmatrix}$$

$$= (\lambda-2)(\lambda-6) - 12 = 0$$

$$= \lambda^2 - 8\lambda + 12 - 12 = 0$$

$$\lambda(\lambda-8) = 0$$

$$\lambda = 0, 8.$$

Yes! the matrix is diagonalizable b/c it's 2×2
w/ 2 distinct e-values, so L is diagonalizable.

(Problem 3 continued) Again, let $L : P_1 \rightarrow P_1$ be a linear transformation with

$$M_B(L) = \begin{bmatrix} 2 & -3 \\ -4 & 6 \end{bmatrix} \text{ with respect to the ordered basis } B = (x+1, x-2).$$

c. (3 points) Is L an isomorphism? Justify your answer.

$$\det(A) = 2 \cdot 6 - (-4)(-3) = 12 - 12 = 0.$$

No. A is not invertible, so L is not an isomorphism.

d. (3 points) Suppose $P_{B \leftarrow D} = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$ is the change matrix from unknown basis $D = (p_1, p_2)$ to B . Find p_1 and p_2 .

$$P_{B \leftarrow D} = [C_B(p_1) \ C_B(p_2)] \quad \begin{aligned} p_1 &= 1(x+1) - 3(x-2) \\ p_2 &= 0(x+1) + 1(x-2) \end{aligned}$$

e. (4 points) For the same basis D as in part (d), find $M_D(L)$.
(Hint: You do NOT need to know D to solve this problem).

$$P_{D \leftarrow B} = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$$

$$M_D(L) = P_{D \leftarrow B} M_B(L) P_{B \leftarrow D}$$

$$= \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 2 & -3 \\ -4 & 6 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 11 & -3 \\ -22 & 6 \end{bmatrix} = \begin{bmatrix} 11 & -3 \\ 11 & -3 \end{bmatrix}$$

4. (8 points total)

Consider the vector space P_1 , and define the function

$$\langle p(x), q(x) \rangle = \int_0^1 p(x)q(x)dx$$

for $p(x), q(x)$ in P_1 .

You can assume that $\langle p(x), q(x) \rangle$ is an inner product for P_1 .

Consider the set $S = \{x+1, 9x-5\}$.

a. (4 points) Show that S forms a basis for P_1 .

$$\text{Lin. indep b/c if } c(x+1) = 9x-5,$$

$$c = 9 \text{ and } c = -5 \text{ X.}$$

Two lin. indep. vectors in a 2-dim'l
vector space are a basis.

b. (4 points) Show that S is an orthogonal set with respect to the inner product $\langle p(x), q(x) \rangle$.

$$\begin{aligned} \langle x+1, 9x-5 \rangle &= \int_0^1 (x+1)(9x-5)dx \\ &= \int_0^1 9x^2 + 4x - 5 dx \\ &= 3x^3 + 2x^2 - 5x \Big|_0^1 = 3 \cdot 1^3 + 2 \cdot 1^2 - 5 \cdot 1 - 0 \\ &= 0 \text{ "}. \\ &\quad \uparrow \text{ means } \perp. \end{aligned}$$

5. (12 points)

You can use the fact that the matrix $\begin{bmatrix} 1 & 0 & 0 & 2 & a \\ 2 & -1 & 0 & 7 & b \\ 3 & -2 & 1 & 13 & c \\ 7 & -4 & 2 & 28 & d \end{bmatrix}$ reduces to $\begin{bmatrix} 1 & 0 & 0 & 2 & a \\ 0 & 1 & 0 & -3 & (2a-b) \\ 0 & 0 & 1 & 1 & (c+a-2b) \\ 0 & 0 & 0 & 0 & (d-a-2c) \end{bmatrix}$ via elementary row operations.

Consider $W = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \\ 7 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ -2 \\ -4 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 7 \\ 13 \\ 28 \end{bmatrix} \right\}$.

a. (3 points) Does $\begin{bmatrix} 1 \\ -2 \\ 2 \\ 4 \end{bmatrix}$ lie in W ? Justify your answer.

$$a=1, b=-2, c=2, d=4 \rightarrow d-a-2c = 4-1-2 \cdot 2 = -1 \neq 0.$$

Last row will be $[0 \ 0 \ 0 \ 0 \ ; \ -1] \leftarrow$ inconsistent.

No.

b. (3 points) Does $\begin{bmatrix} 0 \\ 1 \\ 3 \\ 6 \end{bmatrix}$ lie in W ? Justify your answer.

$$a=0, b=1, c=3$$

$d=6 \rightarrow d-a-2c = 6-0-2 \cdot 3 = 0$. Yes! Last row is $[0 \ 0 \ 0 \ 0 \ ; \ 0]$, every other row has a pivot, so this is a consistent system.

c. (3 points) Is W isomorphic to M_{22} ? Justify your answer in one sentence.

$$\dim(W) = 3 = \# \text{ of pivots.}$$

$$\dim(M_{22}) = 4. \text{ No, } 3 \neq 4.$$

d. (3 points) For which d is W isomorphic to P_d ? Justify your answer in one sentence.

$$3 = d+1 = \dim(P_d) \Rightarrow d=2 \text{ makes dimensions match.}$$

6. (8 points) Consider $T : P_2 \rightarrow P_1$ by $T(p(x)) = p'(x) + xp''(x)$. You can assume that T is a linear transformation.

Find $M_{DB}(T)$, where $B = (x, x^2 + 3x, 1)$ and $D = (1, x)$. Show all work.

(Hint: your final matrix will be 2×3).

$$T(x) = (x)' + x(x)'' = 1 + 0 = 1 \rightarrow \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\begin{aligned} T(x^2 + 3x) &= 2x + 3 + x(2) \\ &= 4x + 3 \rightarrow \begin{bmatrix} 3 \\ 4 \end{bmatrix} \end{aligned}$$

$$T(1) = 0 + x(0) = 0 \rightarrow \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$M_{DB}(T) = \begin{bmatrix} 1 & 3 & 0 \\ 0 & 4 & 0 \end{bmatrix}$$

7. (12 points total) (This problem has overflow room on the next page)

Consider $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ in $\mathbb{R}^2$, and define the function

$$\langle \mathbf{u}, \mathbf{v} \rangle = 3u_1v_1 + 4u_2v_2.$$

a. (4 points) Verify the following inner product property for $\langle \mathbf{u}, \mathbf{v} \rangle$:

$\langle \mathbf{u}, c\mathbf{v} \rangle = c\langle \mathbf{u}, \mathbf{v} \rangle$ for all $\mathbf{u}, \mathbf{v}$ in $\mathbb{R}^2$ and scalar c .

$$\begin{aligned} \left\langle \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, c \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \right\rangle &= \left\langle \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \begin{bmatrix} cv_1 \\ cv_2 \end{bmatrix} \right\rangle = 3u_1(cv_1) + 4u_2(cv_2) \\ &= c(3u_1v_1 + 4u_2v_2) \\ &= c \left\langle \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \right\rangle = c \langle \mathbf{u}, \mathbf{v} \rangle. \end{aligned}$$

For parts (b) and (c), assume the remaining inner product properties are true for

$$\langle \mathbf{u}, \mathbf{v} \rangle = 3u_1v_1 + 4u_2v_2.$$

b. (4 points) Using this inner product $\langle \mathbf{u}, \mathbf{v} \rangle$, verify that the Cauchy-Schwarz¹ inequality

($|\langle \mathbf{v}, \mathbf{w} \rangle| \leq \|\mathbf{v}\| \|\mathbf{w}\|$) holds for $\mathbf{v} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$ and $\mathbf{w} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

$$\begin{aligned} |\langle \mathbf{v}, \mathbf{w} \rangle| &= |3 \cdot 2 \cdot 0 + 4(-1)(1)| \\ &= |-4| = 4. \end{aligned}$$

$$\begin{aligned} \|\mathbf{v}\| &= \sqrt{3 \cdot 2 \cdot 2 + 4(-1)(-1)} \\ &= \sqrt{12 + 4} = \sqrt{16} = 4 \end{aligned}$$

$$4 \leq 4 \cdot 2 = 8 \checkmark$$

$$\|\mathbf{w}\| = \sqrt{3 \cdot 0 + 4 \cdot 1^2} = \sqrt{4} = 2.$$

c. (4 points) Still using this inner product $\langle \mathbf{u}, \mathbf{v} \rangle$, find the distance between $\begin{bmatrix} 2 \\ -3 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Show all work. You do not need to simplify your numerical answer.

$$\begin{bmatrix} 2 \\ -3 \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -5 \end{bmatrix} \leftarrow \text{Find length.}$$

$$\left\| \begin{bmatrix} 1 \\ -5 \end{bmatrix} \right\| = \sqrt{3 \cdot 1^2 + 4(-5)^2} = \sqrt{3 + 100} = \sqrt{103}$$

¹Also called Cauchy-Bunyakovsky-Schwarz

(Overflow room for Problem 7)

8. (11 points total)

- a. (6 points) Suppose $A = \begin{bmatrix} 2 & x \\ -1 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$, and you know that $A+B$ is similar to $\begin{bmatrix} 2 & -1 \\ 3 & 8 \end{bmatrix}$.

Find x .

$$A+B = \begin{bmatrix} 2+1 & x+4 \\ -1+2 & 4+3 \end{bmatrix} = \begin{bmatrix} 3 & x+4 \\ 1 & 7 \end{bmatrix} \quad \leftarrow \text{same determinant.}$$

$$\det(A+B) = 3 \cdot 7 - (x+4) = 2 \cdot 8 - (3)(-1)$$

$16+3=19$

$$21 - x - 4 = 19$$

$$17 - x = 19$$

$$17 - 19 = x$$

$$\boxed{x = -2}$$

- b. (5 points) Suppose C and D are invertible matrices with $C^{-1} = \begin{bmatrix} 4 & -1 \\ 2 & 2 \end{bmatrix}$ and $D^{-1} = \begin{bmatrix} 0 & 2 \\ -1 & 1 \end{bmatrix}$.

If $CDE = \begin{bmatrix} 3 & -4 \\ 0 & 1 \end{bmatrix}$, find E . Show all work.

$$(CD)^{-1} = D^{-1}C^{-1}$$

$$E = \begin{bmatrix} 0 & 2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 4 & -1 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 3 & -4 \\ 0 & 1 \end{bmatrix}$$

$$E = \begin{bmatrix} 4 & 4 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} 3 & -4 \\ 0 & 1 \end{bmatrix}$$

$$E = \begin{bmatrix} 12 & -12 \\ -6 & 11 \end{bmatrix}$$

9. (11 points) Suppose A is a 4×4 matrix with

$$RREF(A) = \begin{matrix} & \begin{matrix} x_1 & x_2 & x_3 & x_4 \end{matrix} \\ \begin{bmatrix} 1 & 0 & -2 & 1 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

This is the reduced row echelon form of A .

a. (2 points) What is the rank of A ? Briefly (one sentence) explain your answer.

rank of A is 2 $\rightarrow$ 2 leading ones/pivots in $RREF(A)$.

b. (4 points) Give a basis for the nullspace of A . Show all work.

$$\begin{aligned} x_3 &= r \\ x_4 &= t \\ x_1 &= 2r - t \\ x_2 &= -3r - 4t \end{aligned} \quad \begin{pmatrix} 2r - t \\ -3r - 4t \\ r \\ t \end{pmatrix} = r \begin{pmatrix} 2 \\ -3 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ -4 \\ 0 \\ 1 \end{pmatrix}$$

Basis

c. (5 points) Give a matrix A with $RREF(A)$ above, where the eigenvalues of A are $\lambda = \text{given}$. Your work should justify that the eigenvalues of A are correct and A reduces to $RREF(A)$. $-3, 2, 0, 0$.

$$\text{let } A = \begin{bmatrix} -3 & 0 & 6 & -3 \\ 0 & 2 & 6 & 8 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[\frac{1}{2}r_2 \rightarrow r_2]{-\frac{1}{3}r_1 \rightarrow r_1} \begin{bmatrix} 1 & 0 & -2 & 1 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = RREF(A) \checkmark$$

E-values:

$$\det(\lambda I - A) = \begin{vmatrix} \lambda + 3 & 0 & -6 & 3 \\ 0 & \lambda - 2 & -6 & -8 \\ 0 & 0 & \lambda & 0 \\ 0 & 0 & 0 & \lambda \end{vmatrix} = (\lambda + 3)(\lambda - 2)\lambda^2 = 0$$

$\lambda = -3, 2, 0, 0 \checkmark$

Scratch paper page! (DO NOT REMOVE-MUST BE TURNED IN)

Back of scratch paper page! (DO NOT REMOVE-MUST BE TURNED IN)